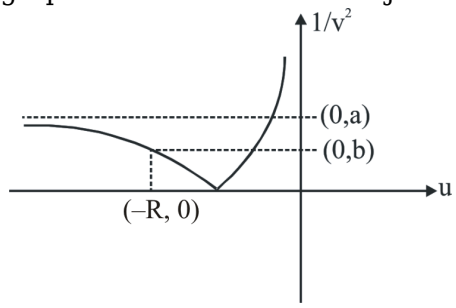
PHYSICS

1) For a concave mirror, graph of $\frac{1}{v^2}$ and u for real object is drawn. What is the value of $\frac{b}{a}$?



(R = radius of curvature)

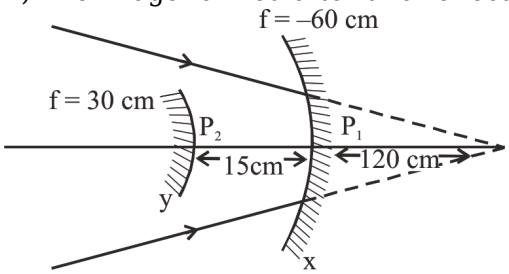
(1) $\frac{1}{4}$

(2) $\frac{1}{2}$

(3) 2

(4) 4

2) The image formed after two reflection (first at x then at y) will be :-



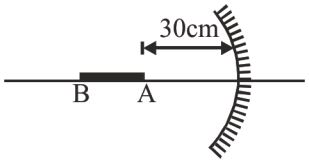
(1) Erect and enlarged 2 times

(2) Inverted and enlarged 2 times

(3) Erect and diminished 2 times

(4) Inverted and diminished 4 times

3) An object AB is placed on the axis of a concave mirror of focal length 10 cm. End A of the object is at 30 cm from the mirror. Find the length of the image. If length of object is 5 cm.



(1) 1 cm

(2) 2 cm

(3) 3 cm

(4) 4 cm

4) The distance of an object from the focus of a convex mirror of radius of curvature 'a' is 'b'. Then the distance of the image from the focus is:

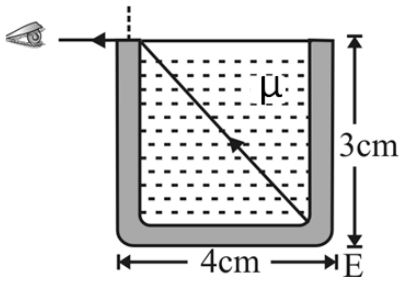
(1) $b^2/4a$

(2) a/b^2

(3) $a^2/4b$

(4) $4b/a^2$

5) When the rectangular metal tank is filled to the top with an unknown liquid. An observer with eyes level with the top of the tank can just see the corner E. A Ray that refracts towards the observer at the top surface of the liquid is shown. The refractive index of the liquid will be :-



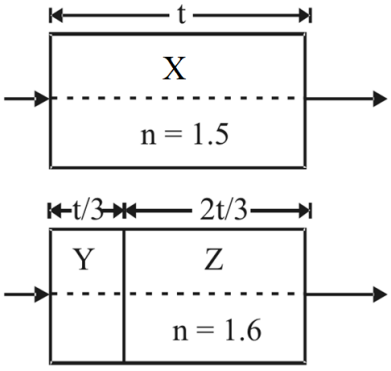
(1) 1.25

(2) 1.40

(3) 1.60

(4) 1.9

6) Two transparent slabs have the same thickness. One is made of material X of refractive index 1.5. The other is made of two material Y and Z having thickness in the ratio 1 : 2. The refractive index of Z is 1.6. If a monochromatic parallel beam passing through the slabs has the same number of wavelengths inside both, the refractive of Y is -



(1) 1.1

(2) 1.2

(3) 1.3

(4) 1.4

7) **Assertion :-** Refractive index of turpentine with respect to air is more than that of water with respect to air.
Reason :- Mass density of turpentine is more than that of water.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

8) A glass slab has a critical angle of 30°, when placed in air. What will be the critical angle, when it is placed in liquid of refractive index 6/5 :-

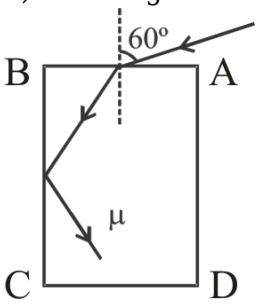
(1) 45°

(2) 37°

(3) 53°

(4) 60°

9) In the figure the minimum value of refractive index if T.I.R. occurs on surface BC :-



(1) 1.732

(2) 1.52

(3) 1.32

(4) 1.12

10) If one face of a prism of prism angle 30° and $\mu = \sqrt{2}$ is silvered, the incident ray retraces its initial path. The angle of incidence is :-

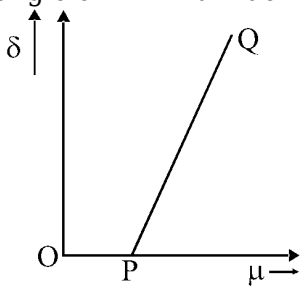
(1) 60°

(2) 30°

(3) 45°

(4) 90°

11) For a small angled prism, angle of prism A, the angle of minimum deviation (δ) varies with the refractive index of the prism as shown in the graph.



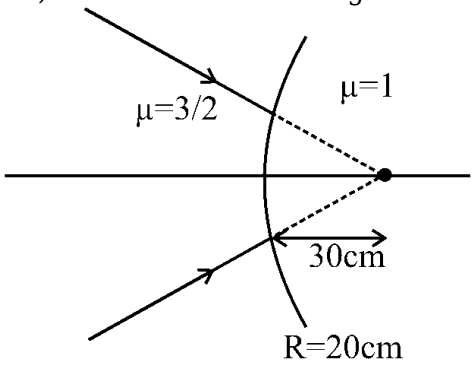
(1) Point P corresponds to μ = 1/2

(2) Slope of the line PQ = A/2

(3) Slope of the line PQ = A

(4) None of the above statements is true

12) After refraction through the curved surface the image of the converging beam is formed at :-



(1) x = 40 cm

(2) x = 40/3 cm

(3) x = – 40/3 cm

(4) x = 20 cm

13) An air bubble is inside water. The refractive index of water is 4/3. At what distance from the air bubble should a point object be placed so as to form a real image at the same distance from the bubble ?

(1) 2R

(2) 3R

(3) 4R

(4) The air bubble cannot form a real image

14) A lens which has focal length of 4 cm and refractive index of 1.4 is immersed in a liquid of refractive index 1.6, then the focal length in liquid will be

(1) – 12.8 cm

(2) 30 cm

(3) 20 cm

(4) – 30 cm

15) An object is placed 12 cm to the left of a diverging lens of focal length 6 cm. A converging lens with a focal length of 12 cm is placed at a distance d to the right of the diverging lens. Find the distance d that corresponds to a final image at infinity.

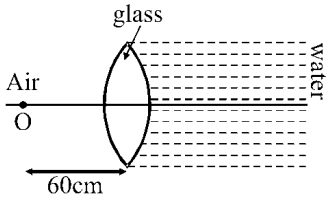
(1) 12 cm

(2) 5 cm

(3) 4 cm

(4) 8 cm

16) If radius of curvature of each surface of given lens is 20 cm. Then position of image of given

object :- 

(1) 80 cm

(2) 160 cm

(3) 40 cm

(4) None of the above

17) ***Assertion*** :- In a simple microscope, image formed is virtual, erect and magnified.
Reason :- Object to be seen is held between optical centre and focus of the convex lens.

(1) Both assertion & reason are true & the reason is a correct explanation of the assertion.

(2) Both assertion & reason are true but reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both assertion & reason are false.

18) In displacement method, the distance between object and screen is 96 cm. The ratio of length of two images formed by a converging lens placed between them is 4, then

(1) Distance between the two positions of the lens is 48 cm

(2) Distance between the two positions of the lens is 32 cm

(3) Focal length of the lens is 64/3 cm

(4) Both (2) & (3) are correct

19) In the displacement method, a convex lens is placed in between an object and a screen. If one of the magnification is 3 and the displacement of the lens between the two positions is 24 cm then the focal length of the lens is :

(1) 10 cm

(2) 9 cm

(3) 6 cm

(4) $\frac{16}{3}$cm

20) In a compound microscope, the focal length of two lenses are 1.5 cm and 6.25 cm. If an object is placed at 2 cm from objective and the final image is formed at 25 cm from eye lens, the distance between two lenses is -

(1) 6.00 cm

(2) 7.75 cm

(3) 9.25 cm

(4) 11.0 cm

21) An astronomical telescope has focal lengths 100 & 10cm of objective and eyepiece lens

respectively when final image is formed at least distance of distinct vision, magnification power of telescope will be :

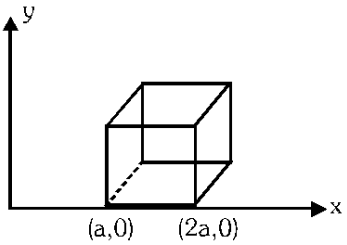
(1) –10

(2) –11

(3) –14

(4) –15

22) As shown in fig. the electric field in a region is given by $E = \frac{E_0}{a^2} x^2 \hat{(i)}$ volt/m then the total electric flux through the cube of side 'a' will be :-



(1) Zero

(2) $3\ E_0\ a^2$

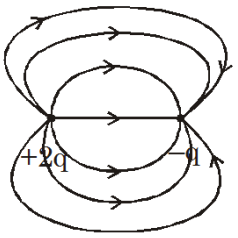
(3) $E_0 a^2$

(4) $3E_0$

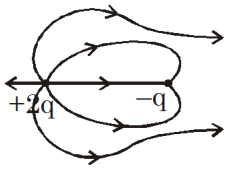
23)

The electric field line pattern for charges +2q & – q are shown. Select the most appropriate diagram :-

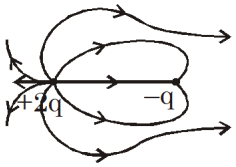
(1)

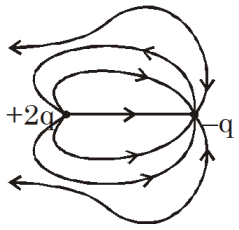


(2)



(3)



(4) 

24) Two point charge of 100 μC and –4 μC are positioned at points $(-2\sqrt{3},\ 3\sqrt{3},\ -4)$ and $(4\sqrt{3},\ -5\sqrt{3},\ 6)$ respectively of a Cartesian coordinate system. Find the force vector on the –4 μC charge? All the coordinates are in meters.

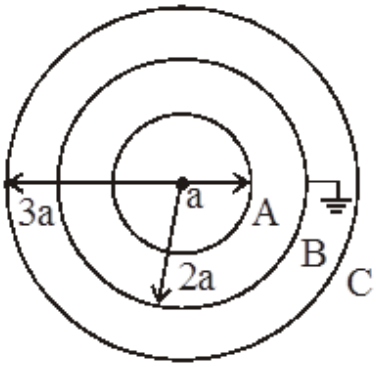
(1) $9 \times 10^{-4}\left(3\sqrt{3}\hat{i} - 4\sqrt{3}\hat{j} + 5\hat{k}\right)$ N

(2) $9 \times 10^{-4}\left(-3\sqrt{3}\hat{i} + 4\sqrt{3}\hat{j} - 5\hat{k}\right)$ N

(3) $2.25 \times 10^{-4}\left(-3\sqrt{3}\hat{i} + 4\sqrt{3}\hat{j} - 5\hat{k}\right)$ N

(4) $2.25 \times 10^{-4}\left(3\sqrt{3}\hat{i} - 4\sqrt{3}\hat{j} + 5\hat{k}\right)$ N

25) Figure shows a system of three concentric metal shells A, B and C with radii a, 2a and 3a respectively. Shell B is earthed and shell C is given a charge Q. Now if shell C is connected to shell A, then the final charge on the shell B, is equal to :-



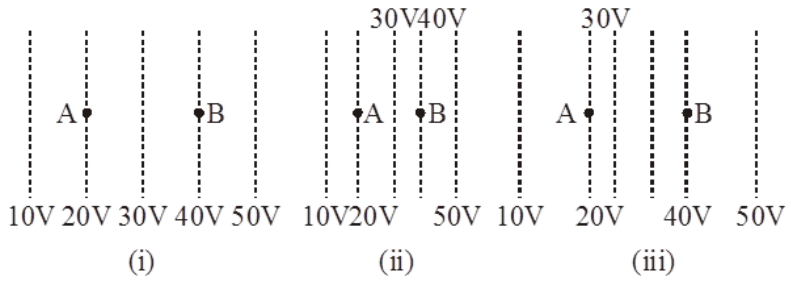
(1) $-\frac{4Q}{13}$

(2) $-\frac{8Q}{11}$

(3) $-\frac{5Q}{3}$

(4) $-\frac{3Q}{7}$

26) Figure shows some equipotential lines distributed in space. A charged object is moved from point A to point B:-



(1) The work done in figure (i) is the greatest.

(2) The work done in figure (ii) is the least.

(3) The work done is the same in figure (i), (ii) and (iii).

(4) The work done in figure (iii) is greater than figure (ii) but equal to that in figure (i).

27) A conducting sphere of radius R is given a charge Q. Consider three points A, B, C such that B at the surface, A at centre and C at a distance R/2 from the centre. The electric potential at these points are such that

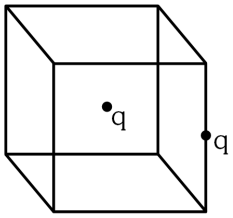
(1) $V_A = V_B = V_C$

(2) $V_A = V_B \neq V_C$

(3) $V_A \neq V_B \neq V_C$

(4) $V_A \neq V_B = V_C$

28) Two charges 'q' each are placed as shown in figure. Then flux passing through the cube is –



(1) $\frac{3}{2}\left(\frac{q}{\varepsilon_0}\right)$

(2) $\frac{5}{4}\left(\frac{q}{\epsilon_0}\right)$

(3) $\frac{2}{3}\left(\frac{q}{\epsilon_0}\right)$

(4) $\frac{5}{12}\left(\frac{q}{\epsilon_0}\right)$

29) Two point charge A and B have charge 0.5 C and 2C respectively and their position vectors are ($\hat{i}+\hat{j}+\hat{k}$) m and ($-\hat{i}-\hat{j}+3\hat{k}$) m respectively. The force on the charge at A due to B is –

(1) $\frac{k(\hat{i}+\hat{j}-\hat{k})}{12\sqrt{3}}$N

(2) $\frac{k}{24\sqrt{3}}(\hat{i}-\hat{j}+\hat{k})$ N

(3) $\frac{k}{24\sqrt{3}}(\hat{i}+\hat{j}+\hat{k})$ N

(4) $\frac{k}{24\sqrt{3}}(-\hat{i}-\hat{j}+\hat{k})$ N

30) A point charge q is placed in a cubical box of side a. If all the dimensions of box are changed by a factor of 3 then by what factor the flux crossing the box will change :-

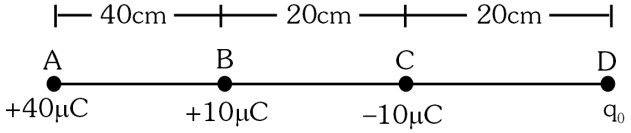
(1) Decrease by 9 times

(2) Increased by 9 times

(3) Either (1) or (2)

(4) Doesn't Change

31) In given diagram, find value of q_0 so that +10μC remain in equilibrium :



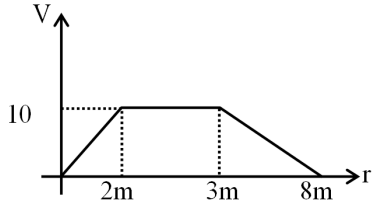
(1) –80 μC

(2) +40 μC

(3) +80 μC

(4) –20 μC

32) In the following graph, the magnitude of maximum electric field is :



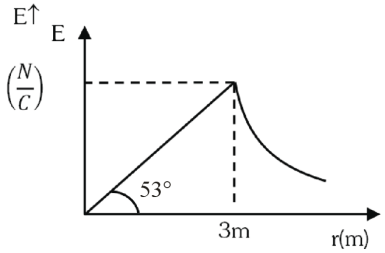
(1) $5Vm^{-1}$

(2) $2.5\ Vm^{-1}$

(3) $1.25\ Vm^{-1}$

(4) $10\ Vm^{-1}$

33) For a solid non – conducting sphere, electric field variation with distance from the centre is shown. Then net charge in the sphere is (In coulomb) –



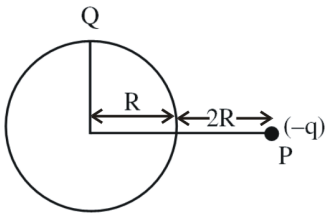
(1) $180\ \pi\in_0$

(2) $162\ \pi\in_0$

(3) $144\ \pi\in_0$

(4) $72\pi\in_0$

34) Electric potential energy of charge (–q) is –



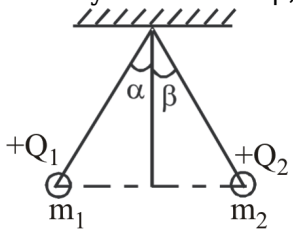
(1) $\frac{kQq}{2R}$

(2) $\frac{kQq}{R}$

(3) $\frac{-kQq}{3R}$

(4) $\frac{kQq}{3R}$

35) Consider the following charged suspended ball system. If $\alpha < \beta$, then the statements which can be true at equillibrium out of followings are :-



(a) $Q_1 > Q_2$, $m_1 < m_2$ (b) $Q_1 > Q_2$, $m_1 > m_2$

(c) $Q_1 < Q_2$, $m_1 = m_2$ (d) $Q_1 < Q_2$, $m_1 > m_2$

(1) Only d

(2) Only c

(3) (c, d)

(4) (b, d)

36) The electric potential at a point (x, y, z) is given by $V = -x^2y - xz^3 + 4$. The electric field $\vec{E}$ at that point is:

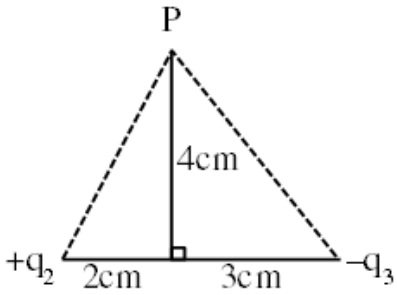
(1) $\vec{E} = \hat{i}2xy + \hat{j}(x^2 + y^2) + \hat{k}(3xz - y^2)$

(2) $\vec{E} = \hat{i}z^3 + \hat{j}xyz + \hat{k}z^2$

(3) $\vec{E} = \hat{i}(2xy - z^3) + \hat{j}xy^2 + \hat{k}3z^2x$

(4) $\vec{E} = \hat{i}(2xy + z^3) + \hat{j}x^2 + \hat{k}3xz^2$

37) If the net electric field at point P along Y axis is zero, then the ratio of $\frac{q_2}{q_3}$ is $\frac{8}{5\sqrt{x}}$, where x = ...



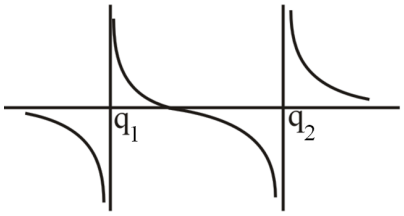
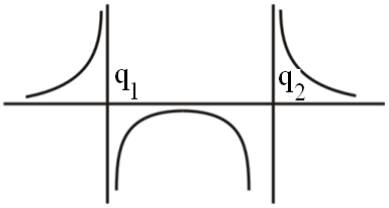
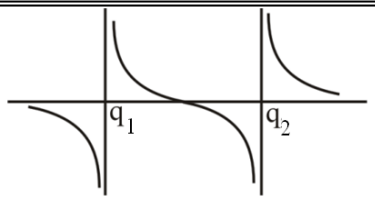
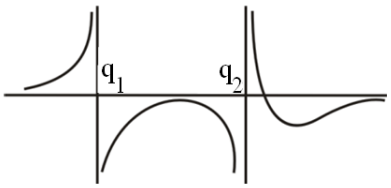
(1) 5

(2) 6

(3) 7

(4) 8

38) Variation of electric field along the line joining two charges is given in list-I. Relation between charges is given in list-II. Match the proper variation. Take electric field along right as positive

List-I		List-II	
(P)		(1)	$\|q_1\| > \|q_2\|$ $q_1 < 0; q_2 > 0$
(Q)		(2)	$\|q_1\| = \|q_2\|$ $q_1 > 0; q_2 > 0$
(R)		(3)	$\|q_1\| = \|q_2\|$ $q_1 < 0; q_2 > 0$
(S)		(4)	$\|q_2\| > \|q_1\|$ $q_1 > 0; q_2 > 0$

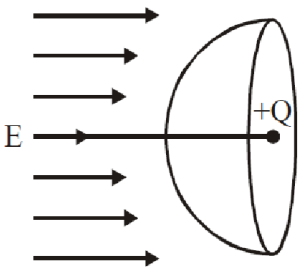
(1) (P) - (3) ; (Q) - (1) ; (R) - (2) ; (S) - (4)

(2) (P) - (1) ; (Q) - (4) ; (R) - (2) ; (S) - (3)

(3) (P) - (4) ; (Q) - (3) ; (R) - (2) ; (S) - (1)

(4) (P) - (2) ; (Q) - (3) ; (R) - (4) ; (S) - (1)

39) The figure shows a point charge +Q placed at the center of an imaginary hemispherical surface of radius R. A uniform electric field E is applied in the horizontal direction. If the net flux passing through the hemispherical surface is zero then what is the magnitude of the electric field E.



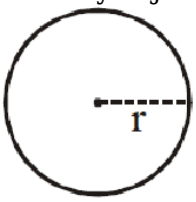
(1) $\frac{Q}{2\pi\varepsilon_0 R^2}$

(2) $\frac{Q}{2\varepsilon_0 R^2}$

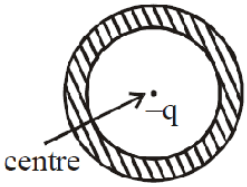
(3) $\frac{Q}{\pi\varepsilon_0 R^2}$

(4) $\frac{Q}{\varepsilon_0 R^2}$

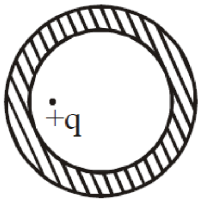
40) In which of the case(s) we will get uniform charge distribution on external spherical surface. Given every object is a conductor.

(A) 

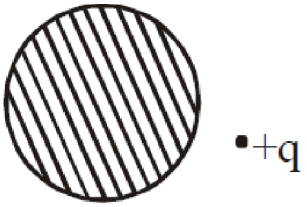
thin shell having net charge +q

(B) 

thick shell having net charge zero

(C) 

thick shell having net charge zero

(D) 

solid sphere having net charge +q

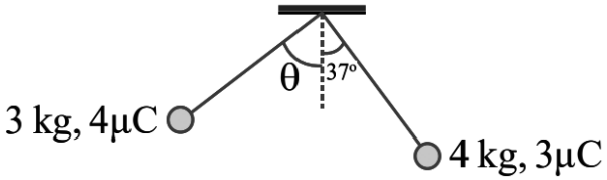
(1) Only (A) & (B)

(2) Only (A), (B) & (C)

(3) Only (A)

(4) All

41) Two small positively charged spheres are suspended from a common point at the ceiling by the insulating light strings of equal length as shown. If one string makes an angle 37° with the vertical,



find the angle θ at equilibrium.

(1) 30°

(2) 37°

(3) 53°

(4) 60°

42) A small sphere carrying a charge 'q' is hanging in between two parallel plates by a string of length L. Time period of pendulum is T_0. When parallel plates are charged, the time period changes to T. The ratio T/T_0 is equal to :

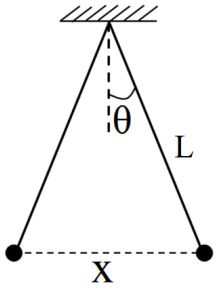
(1) $\left(\frac{g+\frac{qE}{m}}{g}\right)$

(2) $\left(\frac{g}{g+\frac{qE}{m}}\right)^{3/2}$

(3) $\left(\frac{g}{g+\frac{qE}{m}}\right)^{1/2}$

(4) None of these

43) In the given figure two tiny conducting balls of identical mass m and identical charge q hang from non-conducting threads of equal length L. Assume that θ is so small that tanθ = sinθ, then for equilibrium x is equal to :



(1) $\left(\frac{q^2L}{2\pi\varepsilon_0 mg}\right)^{1/3}$

(2) $\left(\frac{qL^2}{2\pi\varepsilon_0 mg}\right)^{1/3}$

(3) $\left(\frac{q^2L^2}{4\pi\varepsilon_0 mg}\right)^{1/3}$

(4) $\left(\frac{q^2L}{4\pi\varepsilon_0 mg}\right)^{1/3}$

44) **Case-1 :** Five identical point charges of magnitude Q are kept at the corners of a regular

pentagon of centre to corner distance a.

Case-2 : One charge is now removed. Match the following for above two cases :

	Column-I		**Column-II**
(A)	Electric field at the centre of pentagon in case-1	(P)	$\frac{1}{4\pi\varepsilon_0}\frac{Q\times 5}{a}$
(B)	Electric potential at the centre of pentagon in case-1	(Q)	Zero
(C)	Electric field at the centre of pentagon in case-2	(R)	$\frac{1}{4\pi\varepsilon_0}\frac{Q}{a^2}$
(D)	Electric potential at the centre of pentagon in case-2	(S)	$\frac{1}{4\pi\varepsilon_0}\frac{Q}{a}\times 4$

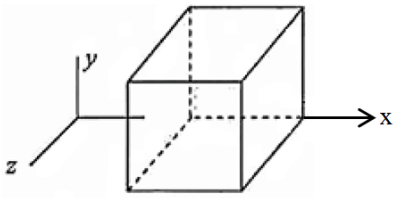
(1) (A → Q); (B → P); (C → R); (D → S)

(2) (A → P); (B → Q); (C → S); (D → R)

(3) (A → S); (B → Q); (C → P); (D → R)

(4) (A → R); (B → Q); (C → P); (D → S)

45) The figure shows a cube of side with edges parallel to the axes. The electric field is in positive y-direction but its magnitude changes such that it is $\frac{15}{\varepsilon_0\ell^2}$ at the bottom face and $\frac{5}{\varepsilon_0\ell^2}$ at the top face in SI unit. Find the charge enclosed by the cube.



(1) 5 C

(2) 10 C

(3) –10 C

(4) Zero

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	**1**	**1**	**1**	**3**	**1**	**3**	**3**	**2**	**3**	**3**	**3**	**1**	**4**	**1**	**4**	**1**	**1**	**4**	**2**	**4**
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	**3**	**2**	**2**	**2**	**2**	**3**	**1**	**2**	**1**	**4**	**3**	**1**	**3**	**3**	**4**	**4**	**1**	**3**	**1**	**2**
Q.	41	42	43	44	45															
A.	**3**	**3**	**1**	**1**	**3**															

SOLUTIONS

PHYSICS

1) From the graph :-
U = – R
$\frac{1}{v^2} = b$
in concave mirror when u = – R
v = – R
$\boxed{b = \frac{1}{R^2}}$
from the graph
$u = -\infty, \frac{1}{v^2} = a$
in concave mirror
u = – ∞ v = – f = R/.2
$\boxed{a = \frac{4}{R^2}}$ then $\frac{b}{a} = \frac{1}{4}$

2) **Question Explanation:** Find the nature and magnification of the final image formed after light rays undergo two successive reflections: first at a concave mirror x (f = –60 cm) and then at a convex mirror y (f = +30 cm), where the mirrors are separated by 15 cm and the initial virtual object for mirror x is at 120 cm.
Concept: Successive Reflections in Spherical Mirrors.
Solution:
For concave mirror
u = +120 cm
f = –60 cm
$v = \frac{uf}{v-f} = \frac{(120)(-60)}{120-(-60)}$ = –40 cm
$m_1 = -\frac{v}{u} = -\frac{-40}{120} = +\frac{1}{3}$
For convex mirror :
u = +(40 – 15) = +25 cm
f = +30 cm
$m_2 = \frac{f}{f-u} = \frac{30}{30-25} = +6$
$m_{net} = m_1 \times m_2 = \frac{1}{3} \times 6 = +2.$
Since m_{net} is positive, the image is erect.
Final answer: Option (1)

3)

for point A
u = – 30 ; f = – 10

$v_1 = \frac{-30(-10)}{-30-(-10)}$ = – 15 cm

for point B

u = – 35 ; f = – 10

$v_2 = \frac{-35(-10)}{-35-(-10)}$ = –14 cm

$L_I = v_2 – v_1$ = – 14 – (– 15) = 1 cm

4) **Explanation:**

The problem asks for the distance of an image from the focus of a convex mirror, given that the object's distance from the focus is b and the radius is a.

Concept:

Newton's Mirror Formula relates distances measured from the focal point.

Formula:

The product of object and image distances from the focus equals the focal length squared: $x \cdot y = f^2$

Where $f = \frac{R}{2}$.

Calculation:

1. **Find Focal Length (f):** Since radius then R = a, then f = a/2.
2. **Identify Distances:** Object distance from focus x = b Let image distance from focus be y.
3. **Substitute into Formula:** $b \cdot y = \left(\frac{a}{2}\right)^2$

$b \cdot y = \frac{a^2}{4}$ Solve for y: $y = \frac{a^2}{4b}$ **Final Answer:** Option (3)

5)

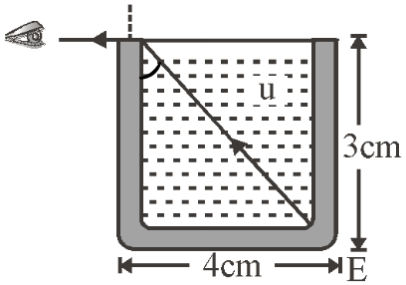
Explain the question : A rectangular tank is filled with unknown liquid; observer at top just sees corner E. Horizontal distance = 4 cm, depth = 3 cm. Ray from E refracts at surface and reaches observer. Find refractive index.

Concept : This question is based on refraction at plane surface (critical case).

Formula : $\mu = 1 / \sin\theta_c$

$\tan\theta_c$ = horizontal distance / depth

Solution :



$\tan\theta_c = 4 / 3$

$\theta_c = \tan^{-1}(4/3) \approx 53.13°$

$\sin\theta_c \approx 3/5$

$\mu = 5/3 = 1.25 \approx 1.2$

Final answer : 1

6)

Wavelength of light

in $X = \frac{\lambda}{1.5}, \text{ in } Y = \frac{\lambda}{\mu}, \text{ in } Z = \frac{\lambda}{1.6}$

λ is wavelength in air

Number of wavelengths in X = number of wavelengths in Y + number of wavelengths in Z

$$\frac{t}{\lambda/1.5} = \frac{t/3}{\lambda/\mu} + \frac{2t/3}{\lambda/1.6}$$

$$1.5 = \frac{\mu}{3} + \frac{3.2}{3}$$

$$\Rightarrow \frac{\mu}{3} = 1.5 - \frac{3.2}{3} = \frac{1.3}{3} \Rightarrow \mu = 1.3$$

7) **Question Explanation:**

We are to check if the refractive index of turpentine > water, and if that is due to turpentine's higher mass density.

Concept Based:

Refractive index depends on optical density, not just mass density. Optical density is related to how much a material slows down light, which depends on electrical polarizability, not directly on mass.

Formula Used:

Refractive index: $\mu = \frac{c}{v}$

No direct relation: μ ∝ mass density

Calculation:

Turpentine has higher refractive index (~ 1.47) than water (~ 1.33) → Assertion is True

Mass density of turpentine (~ 870 kg/m^3) is less than that of water (~ 1000 kg/m^3) → Reason is False

Final Answer:

Option 3 — Assertion is True but the Reason is False.

8) **Explain the Question :** To find the critical angle for glass slab when it in placed in liquid of reproductive index $\frac{6}{5}$.

Concept : This question is based on critical angle and T.I.R.

Solution :

$$\sin i_c = \frac{1}{r\mu d} = \frac{\mu_a}{\mu_g}$$

$$\sin 30° = \frac{1}{\mu_g}$$

$\mu_g = 2$

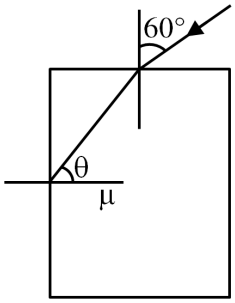
$$\sin i_c = \frac{\mu_i}{\mu_g} = \frac{6/5}{2}$$

$$= \frac{6}{10}$$

$i_c = 37°$
Final Answer : 2

9) **Explain the Question :** To find the minimum value of refractive index if T.I.R. occurs on surface BC **Concept :** This question is based on critical angle and TIR.

Solution : $\sin\theta \leqslant \sqrt{\mu^2 - 1}$



$\theta = 60°$
$\sin^2\theta \leq \mu^2 - 1$
$\sin^2 60° \leq \mu^2 - 1$
$\frac{3}{4} \leqslant \mu^2 - 1$
$\frac{3}{4} + 1 \leqslant \mu^2$
$\mu \geqslant \sqrt{\frac{7}{4}} \Rightarrow \mu \geqslant 1.32$
Final Answer : option (3)

10) **Question Explanation:** For a prism with one silvered face, a light ray retraces its path only if it hits the silvered surface normally. We need to find the incidence angle i that makes this happen.

Concept: Retracing occurs when the refracted ray hits the mirror surface at 90°.

Formula: Condition for retracing: $r_2 = 0°$ which implies $r_1 = A$
Snell's Law: $\mu = \frac{\sin i}{\sin r_1}$

Solution:
Brief Calculation
Identify Values: $A = 30^\circ, \mu = \sqrt{2}$
Refraction Angle: Since the ray hits the second face normally to retrace, $r_2 = 0$ so $r_1 = A = 30°$.
Snell's Law at face 1:

$$\sqrt{2} = \frac{\sin i}{\sin 30^\circ}$$

$$\sin i = \sqrt{2} \times \sin 30^\circ$$

$$\sin i = \sqrt{2} \times \frac{1}{2} = \frac{1}{\sqrt{2}}$$

Solve for $i : i = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45^\circ.$

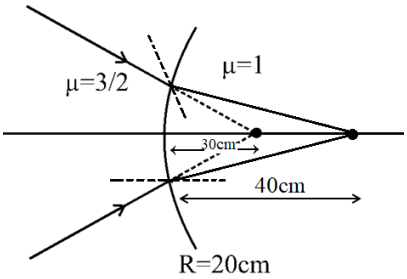
Final Answer: Option (3)

11) $\delta = (\mu - 1)\ A \Rightarrow\ y = (x - 1)\ A = xA - A$
Comparing with y = mx + C
Slope : m = A

12) **Explain question:**
As shown in figure find position of image.

Concept:
Refraction from curved surface.

Solution:
Formula: $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$



$\Rightarrow \frac{1}{v} - \frac{3}{2 \times 30} = \frac{1 - 1.5}{20}$
$\Rightarrow \frac{1}{v} = -\frac{1}{2 \times 20} + \frac{1}{20}$
$\frac{1}{v} = -\frac{1}{40} + \frac{1}{20}$
$\frac{1}{v} = \frac{1}{40}$
$\Rightarrow$ v = 40 cm

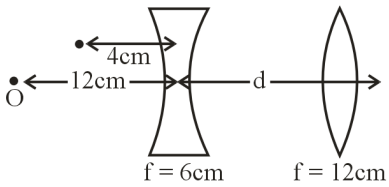
Final answer: Option (1)

13)

Air bubble acts like diverging lens because surrounding medium is denser than inner medium therefore it will diverge the light ray and virtual image is formed.

14) $\frac{1}{f_m} = \left(\frac{\mu_\ell}{\mu_m} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$
$\frac{1}{4} = \left(\frac{1.4}{1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ ...(1)
$\frac{1}{f} = \left(\frac{1.4}{1.6} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ ...(2)
$\frac{(1)}{(2)}\ \ \frac{f}{4} = \frac{0.4 \times 1.6}{-0.2}$

⇒ f = – 12.8 cm



15)

$$\frac{1}{-6} = \frac{1}{v} - \frac{1}{(-12)} \qquad \frac{1}{12} = \frac{1}{\infty} - \frac{1}{-(d+4)}$$

$$\frac{1}{v} = \frac{-2-1}{12} \qquad d + 4 = 12$$

v = –4 cm d = 8 cm

16)

I surface

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{1.5}{v} - \frac{1}{(-60)} = \frac{1.5-1}{20}$$

v = 180 cm

II surface

$$\frac{4/3}{y} - \frac{3/2}{180} = \frac{4/3 - 3/2}{-20}$$

$$\frac{4}{3v} = \frac{1}{120} + \frac{1}{120}$$

$$v = \frac{60 \times 4}{3}$$

= 80 cm

17)

Generated by Allie

Problem Statement: The question asks whether the statement about image formation in a simple microscope is true and if the given reason correctly explains why the image formed is virtual, erect, and magnified.

Underlying Concept: A simple microscope uses a convex lens to magnify objects. When the object is placed between the optical centre and the focal point of the convex lens, the lens forms a virtual, erect, and magnified image of the object. This principle is fundamental in understanding the functioning of magnifying glasses and simple microscopes.

Tips and Tricks: Remember that the simple microscope is essentially a convex lens used as a magnifier - the key is object placement between focal point and optical centre for magnification.

Common Mistakes: Confusing the positions of the object relative to focal point and optical centre; thinking the image formed is real or inverted in this setup.

Why Other Options Are Incorrect?: Answers suggesting the reason is false or not an explanation usually miss that the image nature directly depends on the object's position between focus and optical centre for a convex lens.

18)

Question Explanation:

The question asks you to find the focal length of a converging lens using the displacement method, given the distance between the object and screen (96 cm) and the ratio of the lengths of the two images (4:1) formed by moving the lens between them.

Concept: Displacement Method

Solution: Let object and image distances be x and y,

$\frac{m_1}{m_2} = 4$ and $m_1m_2 = 1 \Rightarrow m_1 = 2$, $m_2 = \frac{1}{2}$ x + y = 96; y/x = 2

y = 2x

3x = 96

x = 32

x + y = 96

x = 32 cm

so y = 64 cm

x–y = 32

Distance between the positions of the lens is 32 cm.

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

⇒ u = – 32, V = 64

put values

$\Rightarrow \frac{1}{64} - \frac{1}{(-32)} = \frac{1}{f}$

$\Rightarrow f = \frac{64}{3} cm$

Focal length of the lens is $\frac{64}{3}$ cm

Final answer : Option (4)

19) Given $M = \frac{v}{u} = 3$

V = 3u

Distance b/w lenses = v – u = 24

3u – u = 24

u = 12 cm and v = 36 cm

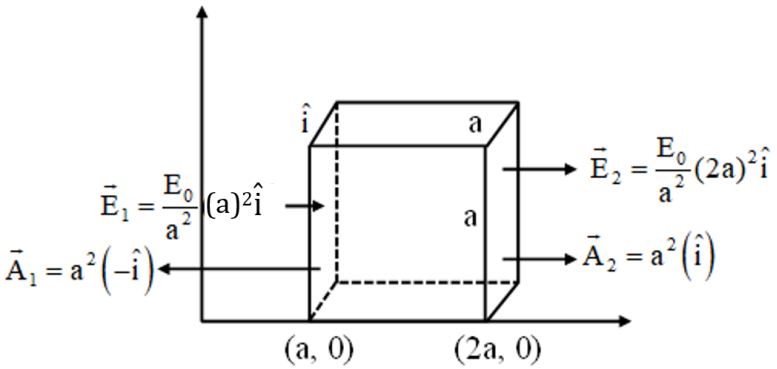
from lens formula

$\frac{1}{f} = \frac{1}{v} - \frac{1}{u} = \frac{1}{36} - \frac{1}{-12} = \frac{4}{36}$

$$f = \frac{36}{4} = 9cm$$

20) Distance between lenses = $|V_0| + |u_e|$
using lens formula for objective,
$|V_0| = 6$ cm
using lens formula for eye piece, $|u_e| = 5$ cm
 Distance between two lenses = 6 + 5 = 11 cm

21) $MP = -\frac{f_0}{fe}\left(1 + \frac{fe}{D}\right)$
$= \frac{-100}{10}\left(1 + \frac{10}{25}\right)$
MP = –14



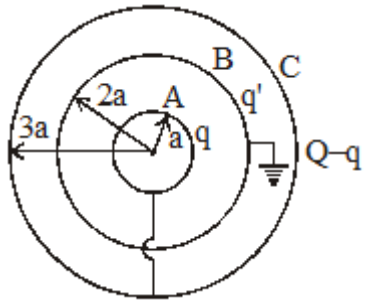
22)
$\phi = \vec{E}_1 \cdot \vec{A}_1 + \vec{E}_2 \cdot \vec{A}_2$
$\phi = E_0(\hat{i}).a^2(-\hat{i}) + \frac{E_0}{a^2}(4a^2)\hat{i}.a^2(\hat{i})$
$\phi = -E_0 a^2 + 4E_0 a^2$
$\phi = 3E_0 a^2$

23)

Field lines are proportional to quantity of charge & they originate at positive charge & terminates at –ve charge or at ∞

24) (1) (2)
100 μc –4μc
$(2\sqrt{3},\ 3\sqrt{3},\ -4)$ $(4\sqrt{3},\ -5\sqrt{3},\ 6)$
$\vec{F}_{21} = \frac{kq_1q_2\,\vec{r}_{12}}{|\vec{r}_{12}|^3}$

25) $V_B = 0$



$$\frac{kq}{2a}+\frac{kq'}{2a}+k\frac{(Q-q)}{3a}=0 \qquad \text{...(i)}$$

$V_A = V_C$

$$\frac{kq}{a}+\frac{kq'}{2a}+\frac{k(Q-q)}{3a}=\frac{k(q+q'+Q-q)}{3a} \qquad \text{...(ii)}$$

from (i) & (ii)

$$q' = \frac{-8Q}{11}$$

26) In all the three figures, V_A = 20 V and V_B = 40 V
Work done in carrying a charge q from A to B is
$W = q\,(V_B - V_A)$
Hence, work done is same in all figure.

27)

In electrostatic condition potential inside a conductor is constant and equal to the potential at the surface.

28)

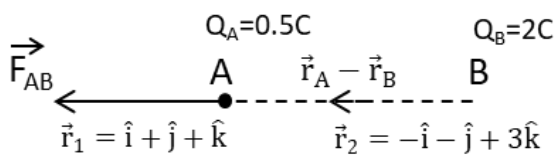
Flux of cube centred charge is $\frac{q}{\epsilon_0}$ and flux of edge centred charge is $\frac{q}{4\epsilon_0}$.
Total flux through the cube

$$= \frac{q}{\epsilon_0}+\frac{q}{4\epsilon_0}=\frac{5}{4}\left(\frac{q}{\epsilon_0}\right)$$

29)

The **vector form of the Coulomb force** on charge **A** due to another point charge **B** is:

$$\vec{F}_{AB}=\frac{kQ_AQ_B}{|r_{AB}|^3}\left[\vec{r}_A-\vec{r}_B\right]$$



$$\vec{F}_{AB}=\frac{K\,(0.5)\,(2)\left[\left(\hat{i}+\hat{j}+\hat{k}\right)-\left(-\hat{i}-\hat{j}+3\hat{k}\right)\right]}{\left|\sqrt{(1-(-1))^2+(1-(-1))^2+(-2)^2}\right|^3}$$

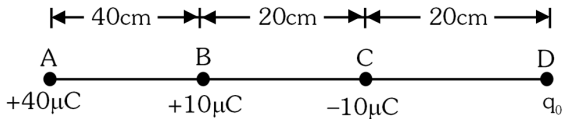
$$\vec{F}_{AB} = \frac{K(1)}{\left[\sqrt{12}\right]^3}\left[2\hat{i} + 2\hat{j} - 2\hat{k}\right]$$

$$\vec{F}_{AB} = \frac{2K}{12 \times 2\sqrt{3}}\left[\hat{i} + \hat{j} - 2\hat{k}\right]$$

$$\vec{F}_{AB} = \frac{K\left(\hat{i} + \hat{j} - \hat{k}\right)}{12\sqrt{3}}$$

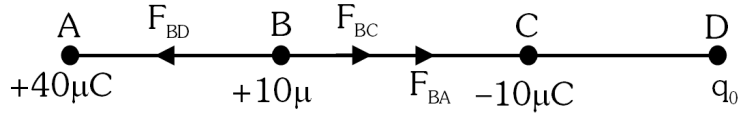
30) Flux does not depend on size.

31)



For equilibrium,

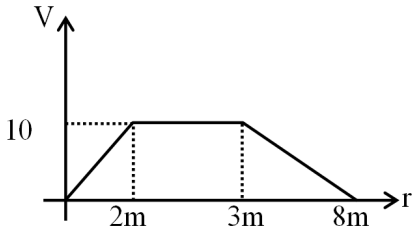
$$F_{BD} = F_{BC} + F_{BA}$$



$$\frac{K10 \times q_0}{(40)^2} = \frac{K10 \times 10}{(20)^2} + \frac{K10 \times 40}{(40)^2}$$

$$\frac{q_0}{(40)^2} = \frac{10}{(20)^2} + \frac{40}{(40)^2}$$

$$q_0 = 10 \times \left(\frac{40}{20}\right)^2 + 40 = +80\ \mu C$$



32)

$$E = -\frac{dv}{dr} \Rightarrow |E| = \left|\frac{dv}{dr}\right| \Rightarrow |E| = (\tan\theta)$$ [slope of V-r curve]

From r = 0 to r = 2m

$$|E_1| = |\tan\theta| = \left|\frac{10}{2}\right| = 5\ \text{V/m}$$

from r = 2m to r = 3m

θ= 0º (line parallel to x-axis)

$$|E_2| = (\tan\theta) = \tan 0^\circ = 0$$

for r = 3 m to r = 8m

$$|E_3| = |\tan\theta| = \left|\frac{10}{5}\right| = 2\ \text{V/m}$$

So the maximum magnitude of E is 5 V/m

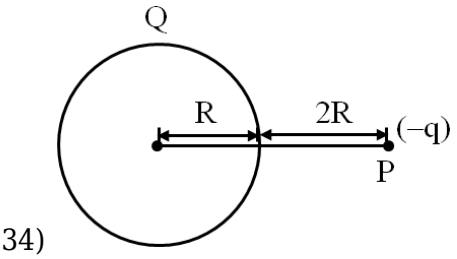
33) Inside a uniformly charged sphere, the linear region has slope $\tan 53° = \frac{4}{3}$. With radius R = 3m, the surface field is $E(R) = \frac{4}{3} \cdot 3 = 4\ N/C$. Using

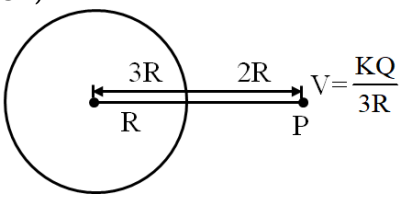
$$E(R) = \frac{1}{4\pi\varepsilon_0 R^2} \Rightarrow Q = 4\pi\varepsilon_0 E(R) R^2$$

we get, substituting E = 4 and R = 3

$Q = 4\pi\varepsilon_0 \cdot 4 \cdot 9 = 144\ \pi\varepsilon_0$ coulomb b

34)



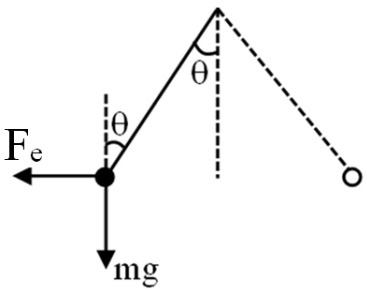


Potential energy at point P = –q × V

$$= -q \times \frac{KQ}{3R}$$

$$= -\frac{KQq}{3R}$$

35)



$$\tan\theta = \frac{F_e}{mg}$$

$$\tan\theta = \frac{KQ_1Q_2}{r^2 mg}$$

$$\theta \propto \frac{1}{m}$$

as $\alpha < \beta \quad m_1 > m_2$

So, $Q_1 > Q_2$ & $m_1 > m_2$

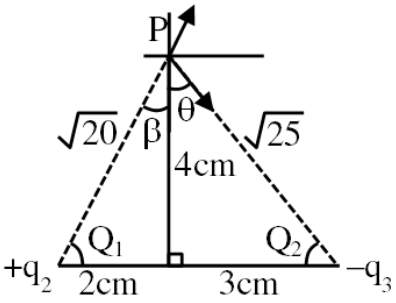
$Q_1 < Q_2$ & $m_1 > m_2$

36)

$$\vec{E} = -\left(\frac{\partial v}{\partial x}\hat{i} + \frac{\partial v}{\partial y}\hat{j} + \frac{\partial v}{\partial z}\hat{k}\right)$$

$$= -\left[\left(-2xy - z^3\right)\hat{i} - x^2\hat{j} - 3xz^2\hat{k}\right]$$

$= \left(2xy + z^3\right)\hat{i} + x^2\hat{j} + 3xz^2\hat{k}$



37)

$$\frac{Kq_2}{20}\cos\beta = \frac{Kq_3}{25}\cos\theta$$

$$\frac{Kq_2}{20}\frac{4}{\sqrt{20}} = \frac{Kq_3}{25}\frac{4}{\sqrt{25}}$$

$$\frac{q_2}{q_3} = \frac{20}{25}\sqrt{\frac{20}{25}} = \frac{8}{5\sqrt{x}}$$

$$\Rightarrow \sqrt{x} = \frac{8 \times 25\sqrt{25}}{5 \times 20\sqrt{20}}$$

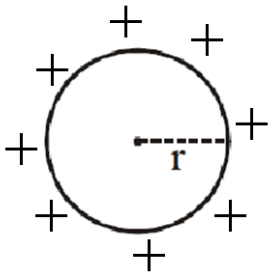
x = 5

38) Apply the following tests :-
(1) Electric field will be zero in between the charges only if the charges are of same nature.
(2) E.F. will be zero in between the charges, closer to the smaller charge
(3) E.F. may be zero on outer side of charges (on the line joining them), if they are unequal, closer to the smaller charge.

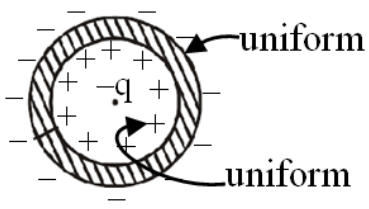
39) $E \times (\pi R^2) = \frac{Q}{2\varepsilon_0}$

$$E = \frac{Q}{2\pi R^2 \varepsilon_0}$$

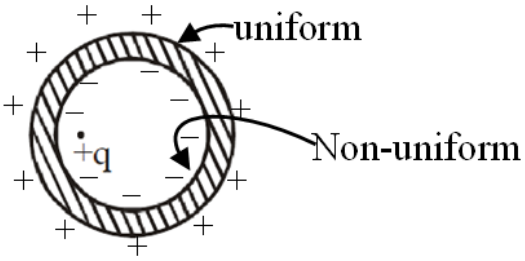
40)



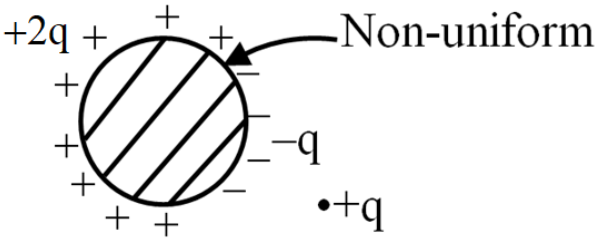
(A)
+q charge is uniformly distributed at outer surface



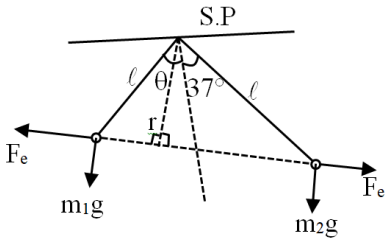
(B)



(C)



(D)



41)

Here system is in equilibrium masses are not equal so line joining masses is not horizontal or angle made by strings with vertical are different. Here net torque on m_1 and m_2 is zero. Here we draw a perpendicular from suspension point to line joining charges. Length of this perpendicular consider as r. Let magnitude of electrostatic force between charge particles is F_e. Now balance the torque on m_1 about suspension point (S.P.)

$F_e\, r = m_1 g \sin\theta$ ...(1)

for m_2 $F_e\, r = m_2 g \sin 37°$ ...(2)

by eq^n (1) & (2)

$$\sin\theta = \frac{m_2}{m_1}\sin 37^\circ = \frac{4}{3}\times\frac{3}{5} = \frac{4}{5}$$

Q = 53°

42)

$$T_0 = 2\pi\sqrt{\frac{\ell}{g}}$$

$$T = 2\pi\sqrt{\frac{\ell}{g_{eff}}} = 2\pi\sqrt{\frac{\ell}{\left(g+\frac{qE}{m}\right)}}$$

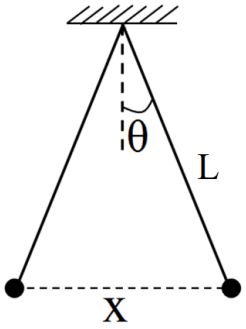
$$\frac{T}{T_0} = \left(\frac{g}{g + \frac{qE}{m}}\right)^{\frac{1}{2}}$$

43) $\tan\theta = \frac{Fe}{mg}$ $\because \theta << \Rightarrow \tan\theta \approx \sin\theta$

$$x^3 = \frac{2kq^2L}{mg}$$

$$x = \left(\frac{2kq^2L}{mg}\right)^{1/3}$$

$\sin\theta = \frac{kq^2}{x^2mg} = \frac{x}{2L}$



44) Point charges are uniformly distributed around the centre O. Hence electric field E is zero.

 (A) → Q.

Electric potential $V = 5 \times \left(\frac{Q}{4\pi\varepsilon_0 a}\right)$

 (B) → P

In case 2, the electric field of three identically placed charges is zero. Net field at the centre of O is due to a single point charge and is given as

$$E = \frac{Q}{4\pi\varepsilon_0 a^2}$$

 (C) → R

In case 2, the electric potential, being scalar, becomes

$$V = 4 \times \frac{Q}{4\pi\varepsilon_0 a}$$

 (D) → S

45) Flux enclosed by the loop,

$$\phi = \frac{-15}{\varepsilon_0 \ell^2}(\ell^2) + \frac{5}{\varepsilon_0 \ell^2}(\ell^2) = \frac{-10}{\varepsilon_0} = \frac{Q_{in}}{\varepsilon_0}$$

So, Q_{in} = –10 C